

# Structural Self-Energy Expansion as a Two-Axiom Cosmology: Algebraic Derivation of the CMB Background from $\varphi$ and $\pi$

SSEE — Paper 4 (Peer-Review Corrected)

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## Abstract

We show that the cosmological background of the observable universe can be described by a two-axiom algebraic system built on the golden ratio  $\varphi = (1 + \sqrt{5})/2$  and the circle constant  $\pi$ . A third structural constant,  $\Omega = \varphi + \pi$ , follows as a theorem rather than an independent axiom; no free parameter is introduced at the background level. Phenomenological extensions (notably the neutrino mass sum  $\Sigma m_\nu$ , classified as Type P later in the paper) were tracked as open problem OP-14 in `OPEN_PROBLEMS.md`, resolved on 2026-06-04. From these two axioms alone we derive, without fitting:

- (i)  $H_0/(\text{km s}^{-1} \text{Mpc}^{-1}) = 3(\varphi + \pi)^2 = 67.962137$  ( $1.1\sigma$  from Planck 2018), where  $M_v = 3\Omega$  is the maximal structural invariant of the dictionary, reached by twenty distinct additive routes under the no-self-sum construction law (Appendix A); this multiplicity is consistent with, but does not by itself prove, the absence of post-hoc tuning;
- (ii) the physical baryon density  $\Omega_b h^2 = (\pi - \varphi)/(3\Omega^2) = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$  ( $0.32\sigma$  from Planck 2018);
- (iii) the primordial spectral index  $n_s = 1 - \varphi^{-7} = 0.965558$  ( $0.16\sigma$  from Planck 2018), where the exponent 7 equals the number of primary structural registers in the SSEE hierarchy;
- (iv) a local Hubble constant from the screening mechanism of Paper 9,  $H_0^{\text{glob}} = H_0^{\text{SHOES}}(1 - f_{\text{scr}}) = 68.13 \text{ km s}^{-1} \text{Mpc}^{-1}$  ( $0.17\sigma$  from the pure algebraic number  $3(\varphi + \pi)^2 = 67.962137$ ), taking [Riess et al. \[2022\]](#) as the input (2026 reframe); the earlier CMB-mapped register (via  $H_0^{\text{MIRA}} = 67.037$ ) is superseded; see Section 6).

These background predictions are number-to-number comparisons with zero adjustable parameters in their respective sectors. The same two axioms fix the cosmological background — including the dark-energy parameters  $w_0 = -0.839950$ ,  $w_a = -0.669975$  derived in the companion Papers 1–3. In the static (Eckart) regime  $\Omega_c h^2 = K_{\text{AL}} \Omega_b h^2 = 0.1235$  ( $+2.9\sigma$ ); modulated by the IS inflationary factor  $n_s = 1 - \varphi^{-7}$  this becomes  $\Omega_c h^2 = K_{\text{AL}} \Omega_b h^2 n_s = 0.1193$  ( $-0.6\sigma$  from

Planck), within  $1\sigma$  of the observed value. A falsifiable prediction  $r = \varphi^{-10} = 0.008131$  is presented for the LiteBIRD satellite mission ( $\sim 2032$ ).

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# 1 Introduction

The discovery of accelerated cosmic expansion [Riess et al., 1998, Perlmutter et al., 1999] and subsequent precision measurements [Planck Collaboration, 2020a] have established the  $\Lambda$ CDM concordance model, which requires six free parameters to describe the CMB power spectrum [Hu et al., 1997]. Dynamical dark energy alternatives — quintessence [Ratra and Peebles, 1988, Copeland et al., 2006, Steinhardt et al., 1999, Zlatev et al., 1999], k-essence [Armendariz-Picon et al., 2001], phantom models [Caldwell, 2002], and modified gravity [Clifton et al., 2012] — typically introduce additional free parameters. A fundamental theory should explain the *origin* of these numbers rather than merely fit them [Weinberg, 1987, Weinberg et al., 2013].

The SSEE framework (Structural Self-Energy Expansion, [Almeida Vallejo, 2026a]) derives the dark-energy equation-of-state parameters  $(w_0, w_a) = (-0.840, -0.670)$  in the Chevallier–Polarski–Linder parametrization [Chevallier and Polarski, 2001, Linder, 2003] algebraically from  $\varphi$  and  $\pi$ , with zero free parameters in that sector [Almeida Vallejo, 2026b,c,d]. Here we extend this programme to the entire cosmological background. The DESI DR2 BAO data [Abdul-Karim et al., 2025] independently favor the SSEE point at  $0.24\sigma$  (Pantheon+;  $0.2\text{--}1.8\sigma$  across SN compilations) in the  $w_0\text{--}w_a$  plane [Almeida Vallejo, 2026c].

**Peer-review notice.** An internal audit identified four vulnerabilities in a prior draft of this paper: (A) use of dimensional quantities without unit justification; (B-1) treating  $\Omega = \varphi + \pi$  as an axiom rather than a theorem; (B-2) post-hoc selection of the exponent in  $n_s = 1 - \varphi^{-7}$ ; (B-3) unexplained factor of 100 in the baryon density. This revised manuscript addresses each point explicitly.

## 2 The SSEE Two-Axiom System

### 2.1 Axioms

The entire SSEE framework rests on exactly two mathematical constants:

$$\varphi = \frac{1 + \sqrt{5}}{2} = 1.618033988\dots \quad (\text{golden ratio}) \quad (1)$$

$$\pi = 3.141592653\dots \quad (\text{circle constant}) \quad (2)$$

Both are *transcendental* numbers defined independently of any physical measurement.

### 2.2 Theorem 1: $\Omega = \varphi + \pi$

The SSEE structural constant  $\Omega$  is **not** an independent axiom. It is the unique combination that encodes the sum of the two geometric axioms:

$$\Omega \equiv \varphi + \pi = 4.759626642\dots \quad (3)$$

This single expression generates the entire hierarchy of SSEE constants (Table 1) through algebraic operations alone.

### 2.3 The Maximal Structural Invariant $M_v$

The value

$$M_v \equiv \varphi + \pi + K_v = 3\Omega = 14.278879927\dots \quad (4)$$

is the *maximal structural invariant* of the constant dictionary. Under the additive construction law—each constant is a sum of distinct previously defined parents, with no constant added to itself—it is the closure value at which the dictionary terminates, coinciding through  $M_v = 3\Omega$  with the copy-law ceiling.

A direct enumeration under this law shows that  $M_v$  is reached by *twenty* distinct additive routes built from the named constants: six using two constants, eleven using three, and three using four (Appendix A). Representative routes, in neutral notation, are

$$\Omega + K_v, \quad \beta + \text{KAL} + P_{sc}, \quad P_{sc} + \Omega + \pi, \quad \varphi + \pi + K_v, \quad (5)$$

each equal to  $M_v$  to 16 significant digits. This multiplicity is a consequence of  $M_v$  being the maximal invariant of the two-dimensional lattice spanned by  $\varphi$  and  $\pi$ ; it is consistent with, but does not by itself prove, the absence of post-hoc tuning.

Table 1: SSEE structural constants. All entries are algebraic in  $\varphi$  and  $\pi$  only; no empirical input is used.

| Name                         | Formula in $\varphi, \pi$              | Simplified               | Value     |
|------------------------------|----------------------------------------|--------------------------|-----------|
| $\Omega$                     | $\varphi + \pi$                        | —                        | 4.759627  |
| $\beta$                      | $(\varphi + \pi)/2$                    | $\Omega/2$               | 2.379813  |
| AURA                         | $(3\varphi + \pi)/2$                   | —                        | 3.997847  |
| MIRA                         | $(3\varphi + \pi)/4$                   | AURA/2                   | 1.998924  |
| KAL                          | $(\varphi + 3\pi)/2$                   | —                        | 5.521406  |
| $P_{sc}$                     | $2\varphi + \pi$                       | $\Omega + \varphi$       | 6.377661  |
| $\Omega + \pi$               | $\varphi + 2\pi$                       | —                        | 7.901219  |
| $K_v$                        | $2\varphi + 2\pi$                      | $\varphi + \pi + \Omega$ | 9.519253  |
| $\pi + \text{KAL}$           | $(\varphi + 5\pi)/2$                   | —                        | 8.663001  |
| $\varphi + \pi + \text{KAL}$ | $(3\varphi + 5\pi)/2$                  | —                        | 10.281034 |
| $2\Omega + \varphi$          | $3\varphi + 2\pi$                      | —                        | 11.137287 |
| $\Omega_b h^2$ (BBN)         | $(\pi - \varphi)/[3(\varphi + \pi)^2]$ | —                        | 0.022420  |
| $M_v$                        | $3(\varphi + \pi)$                     | $\varphi + \pi + K_v$    | 14.278880 |

## 3 Dimensional Analysis: Honest Statement of Scope

A rigorous peer review must address the following question: *How can dimensionless numbers derived from  $\varphi$  and  $\pi$  produce physical quantities with units such as  $\text{km s}^{-1} \text{Mpc}^{-1}$ ?*

**Answer.** They cannot directly; consequently, SSEE treats these relations only as dimensionless numerical correspondences rather than literal derivations of physical units. What SSEE claims is strictly the following:

## Scope of SSEE Predictions

The SSEE system produces **dimensionless numbers** from  $\varphi$  and  $\pi$  that are **numerically equal** to the conventional values of cosmological parameters when those parameters are expressed in the unit systems adopted by the observational community ( $\text{km s}^{-1} \text{Mpc}^{-1}$  for  $H_0$ ; dimensionless density ratios  $\Omega_x h^2$  for matter densities). The claim is not “units are derived”; the claim is “the numerical values are predicted”.

### 3.1 Internal Consistency of the Cosmological Predictions

A single dimensionless coincidence could be dismissed as numerology. What is harder to dismiss is a *system* of mutually consistent cosmological predictions —  $H_0$ ,  $\Omega_b$ ,  $n_s$ , and the Hubble-tension scale — derived from the *same two constants* with *zero adjustments*, assessed against the closed-dictionary look-elsewhere analysis of Paper 1.

Table 2 collects these predictions.

Table 2: Cross-domain predictions of the SSEE two-axiom system. All entries use only  $\varphi$  and  $\pi$ ; no domain-specific parameters are introduced.

| Domain                         | SSEE formula                                        | SSEE value | Observed             | Deviation    |
|--------------------------------|-----------------------------------------------------|------------|----------------------|--------------|
| <b>Cosmology</b>               |                                                     |            |                      |              |
| $H_0$ [km/s/Mpc]               | $3(\varphi + \pi)^2$                                | 67.962137  | $67.4 \pm 0.5$       | $1.1\sigma$  |
| $H_0^{\text{glob}}$ [km/s/Mpc] | $H_0^{\text{SHOES}}(1 - f_{\text{scr}})$            | 68.13      | $3(\varphi + \pi)^2$ | $0.17\sigma$ |
| $\Omega_b$                     | $\frac{100(\pi - \varphi)}{6(\varphi + \pi)^4}$     | 0.0495     | $0.0493 \pm 0.0004$  | $0.53\sigma$ |
| $n_s$ (inflation)              | $1 - \varphi^{-7}$                                  | 0.965558   | $0.9649 \pm 0.0042$  | $0.16\sigma$ |
| <b>Hubble tension</b>          |                                                     |            |                      |              |
| Fractional correction          | $K/H_0 = \frac{\varphi + 3\pi}{6(\varphi + \pi)^2}$ | 8.12%      | $\approx 8\%$        | —            |

<sup>P</sup> Canonical IR-cascade value (2026 reframe):  $f_{\text{scr}}$  applied to the measured  $H_0^{\text{SHOES}} = 73.04$  returns 68.13, i.e.  $0.17\sigma$  from  $3(\varphi + \pi)^2 = 67.962137$ ; the full UV cascade closes to 67.96214 (Paper 10). The earlier CMB-mapped register ( $H_0^{\text{MIRA}} = 67.037$ ) is superseded.  
Consistent with Paper 1 / Paper 9 policy (single anchor shared by every sector).

These cosmological ratios are evaluated against a fixed, closed algebraic dictionary built from  $\{\varphi, \pi\}$ ; a quantitative look-elsewhere analysis of that dictionary is given in Paper 1 and the consolidated dark-energy document, which is the proper basis for assessing global significance rather than an informal product of individual deviations.

## 4 Algebraic Derivation of Cosmological Parameters

### 4.1 Hubble Constant $H_0$

From Section 2.3, the maximal structural invariant is  $M_v = 3(\varphi + \pi)$ . The Hubble constant is the product of this convergent sum and the base unit  $\Omega = \varphi + \pi$ :

$$\frac{H_0}{\text{km s}^{-1} \text{Mpc}^{-1}} = M_v \times \Omega = 3(\varphi + \pi)^2 = 67.962137 \quad (6)$$

**Dimensional note.** The number 67.962137 is compared to  $H_0$  as measured in  $\text{km s}^{-1} \text{Mpc}^{-1}$ . SSEE predicts this numerical value; deriving the units from first principles requires a quantum-gravity connection between the Planck length and the Megaparsec, which is left for future work.

**Algebraic prediction vs. observational posterior.** The value  $H_0^{\text{alg}} = 67.962137$  is a pure geometric prediction. Baryon acoustic oscillations [Eisenstein et al., 2005] provide a standard ruler; independent Bayesian inference against DESI DR2 BAO data [Abdul-Karim

et al., 2025] yields  $H_0^{\text{MCMC}} = 67.79 \pm 0.35 \text{ km s}^{-1} \text{ Mpc}^{-1}$  (Paper 2, under the  $\omega_m$ -direct CMB anchor as prior,  $H_0^{\text{alg}} = 67.962137$ , Paper 3). With the total gravitating matter density  $\Omega_m = \omega_m/h^2 = 0.308881$  in the background geometry, the BAO-corrected posterior is consistent with the algebraic anchor at  $0.50\sigma$ , and the SSEE sound horizon matches  $\Lambda\text{CDM}$  ( $r_d^{\text{SSEE}} = 148.2 \text{ Mpc}$  vs.  $r_d^{\Lambda\text{CDM}} = 147.8 \text{ Mpc}$ ): DESI DR2 is consistent with the algebraic value without singling it out. The  $\sim 0.8 \text{ km s}^{-1} \text{ Mpc}^{-1}$  downshift reported in earlier drafts was an artefact of using the cold sector  $1 + w_0 = 0.160050$  as the background density, now corrected.  $H_0^{\text{alg}}$  is thus the theoretical prediction of the SSEE background;  $H_0^{\text{MCMC}}$  is the same quantity after the observational calibration correction is applied, and the two agree to  $0.50\sigma$ .

## 4.2 Physical Baryon Density $\Omega_b$

The SSEE algebraic structure yields the physical baryon density directly from the CP asymmetry  $(\pi - \varphi)$  normalized to the pure algebraic combination  $3\Omega^2 = 3(\varphi + \pi)^2$ , whose value read in  $\text{km s}^{-1} \text{ Mpc}^{-1}$  is the SSEE Hubble anchor:

$$\Omega_b h^2 = \frac{\pi - \varphi}{3\Omega^2} = \frac{\pi - \varphi}{3(\varphi + \pi)^2} = 0.022418 \quad (7)$$

$$\Omega_b = \frac{\Omega_b h^2}{h^2} = \frac{100(\pi - \varphi)}{6(\varphi + \pi)^4} = 0.04948 \quad (8)$$

Planck 2018:  $\Omega_b h^2 = 0.02237 \pm 0.00015$ ; deviation =  $0.32\sigma$ .

**Sakharov-mechanism motivation (not a quantitative derivation).** We are explicit about the status of Eq. (7): it is an algebraic ansatz whose *form* is motivated by — but not derived from — the Sakharov conditions for baryogenesis [Sakharov, 1967]. The three conditions are realised structurally in SSEE as follows:

- **B-violation:** sphaleron transitions at  $T_{\text{EW}} \sim \Omega$  (SSEE units), with rate  $\Gamma_{\text{sph}} \propto (\alpha_W T)^4 e^{-E_{\text{sph}}/T}$ .
- **CP-violation:**  $\delta_{\text{CP}} = (\pi - \varphi)/\Omega$ , the ratio of the transcendental excess ( $\pi$ , gauge-boson loops) to the algebraic scalar sector ( $\varphi$ ), normalised by  $\Omega = \varphi + \pi$ .
- **Non-equilibrium:** gravitational reheating with  $T_{\text{rh}} \sim 10^{-4} \text{ GeV} \ll T_{\text{EW}}$  ensures departure from equilibrium.

The Sakharov scaling  $\eta_B \propto \delta_{\text{CP}}/(\text{expansion rate})$  motivates the *shape*  $\Omega_b h^2 \sim (\pi - \varphi)/(3\Omega^2)$ , with  $(\pi - \varphi)$  as the CP-asymmetry numerator and  $3\Omega^2$  as the normalising denominator. Two caveats are stated openly. First,  $3\Omega^2$  enters as the dimensionless value 67.962137, i.e.  $H_0^{\text{SSEE}}$  expressed in  $\text{km s}^{-1} \text{ Mpc}^{-1}$ ; this is a Type-P numerical coincidence in the sense of Postulate D of Paper 1, not a dimensional identity. Second,  $\eta_B$  and  $\Omega_b h^2$  differ by the standard photon-entropy factor ( $\eta_B \simeq 2.73 \times 10^{-8} \Omega_b h^2$ ), so the proportionality fixes the algebraic *form* only, not the absolute normalisation. A scan of 96 algebraic candidates (see `ssee_op1.baryon_density.py`) shows Eq. (7) is the unique SSEE monomial within  $1\sigma$  of Planck — a selection statement, not a dynamical derivation. The genuine quantitative test — integrating the Boltzmann equation with the sphaleron rate and the SSEE reheating history to obtain  $\eta_B$  *ab initio* — is deferred to Paper B; the companion framework script `ssee_op1.baryogenesis.py` currently only verifies that the required entropy-dilution factor is consistent with a quintessential-reheating temperature  $T_{\text{rh}} \sim 10^{-4} \text{ GeV}$ , which is a consistency check, not a prediction of  $\eta_B$ . The agreement  $\Omega_b^{\text{SSEE}} = 0.0495$  vs Planck’s  $0.0493 \pm 0.0004$  is  $0.53\sigma$ .

### 4.3 Total Matter Density $\Omega_m$

The CMB-sector *total* matter density is the standard physical observable, built piece-by-piece from SSEE algebra ( $\omega_m$ -direct, OP-8 dissolved):

$$\Omega_{m,\text{CMB}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2} = \frac{0.14267}{(0.67962)^2} = 0.308881, \quad (9)$$

with  $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2]$  and  $\omega_c = K_{\text{AL}_0} \omega_b n_s$  (forward; Paper 3), in  $0.88\sigma$  agreement with Planck 2018 ( $\Omega_m = 0.3153 \pm 0.007$ ). There is one matter density,  $\Omega_m = \omega_m/h^2 = 0.308881$  (no matter factor; OP-8 dissolved; the former Two- $\Omega_m$  criterion is withdrawn, cf. Paper 1, Sec. 1.4). The quantity  $s_m = (\pi - \varphi)/[2(\varphi + \pi)] \approx 0.160$  ( $= 1 + w_0$ ) is an equation-of-state-sector identity, not a density. (The earlier MIRA mapping  $\text{MIRA} \times s_m = 0.31993$  and the geometric route  $(\pi - \varphi)/(\pi + \varphi) = 0.3201$  are both superseded.) The cold dark matter density can be related to the baryon density through the structural retention  $K_{\text{AL}} = \beta + \pi \approx 5.5214$ :

$$\Omega_c h^2 = K_{\text{AL}} \Omega_b h^2 = 0.12351 \quad (+2.9\sigma \text{ Eckart static}). \quad (10)$$

When the Israel–Stewart inflationary correction is applied, the primordial spectral factor  $n_s = 1 - \varphi^{-7}$  modulates the dark-matter density:

$$\boxed{\Omega_c h^2 = K_{\text{AL}} \Omega_b h^2 n_s = 5.521406 \times 0.02242 \times 0.96556 = 0.119514} \quad (11)$$

**Note on  $\Omega_b h^2$  input.** Both the static (Eq. 10) and IS formulas use the Planck 2018 *observed* value  $\Omega_b h^2 = 0.02237 \pm 0.00015$ . The SSEE algebraic prediction  $\Omega_b h^2 = (\pi - \varphi)/(3\Omega^2) = 0.022418$  ( $0.32\sigma$  from Planck) can also be substituted: the IS formula gives  $K_{\text{AL}} \times 0.02242 \times n_s = 0.11952$  ( $-0.4\sigma$ ), consistent with Planck. For transparency, the baseline calculation uses the Planck observed value.

Planck 2018:  $0.1200 \pm 0.0012$ ; deviation =  $-0.6\sigma$  (using Planck observed  $\Omega_b h^2$ ). The IS modulation reduces the tension from  $+2.9\sigma$  (Eckart) to  $-0.6\sigma$  (IS), within  $1\sigma$  of the observed value and consistent with zero tension at  $1\sigma$ .

### 4.4 Primordial Spectral Index and the Seven-Level Hierarchy

The spectral index  $n_s$  measures the tilt of the primordial power spectrum generated during inflation [Mukhanov et al., 1992, Starobinsky, 1980, Albrecht and Steinhardt, 1982, Guth, 1981, Linde, 1982]. Current observational constraints from Planck are  $n_s = 0.9649 \pm 0.0042$  [Planck Collaboration, 2020b].

**Independent justification of the exponent 7.** The SSEE system possesses exactly seven primary structural registers that can be derived sequentially from  $\varphi$  and  $\pi$  without redundancy (Table 3).

Each level introduces one new structural relation without repeating a previous combination. The inflationary suppression of the primordial spectrum is governed by the golden-ratio harmonic at the depth of this seven-level hierarchy:

$$\boxed{n_s = 1 - \varphi^{-7} = 1 - \frac{1}{\varphi^7} = 0.965558} \quad (12)$$

Table 3: The seven primary structural registers of SSEE. Each level is derived from the previous; none is adjustable.

| Level | Register  | Formula              | Value      |
|-------|-----------|----------------------|------------|
| 1     | $\varphi$ | $(1 + \sqrt{5})/2$   | 1.61803... |
| 2     | $\pi$     | (circle constant)    | 3.14159... |
| 3     | $\beta$   | $(\varphi + \pi)/2$  | 2.37981... |
| 4     | $\Omega$  | $\varphi + \pi$      | 4.75963... |
| 5     | AURA      | $(3\varphi + \pi)/2$ | 3.99785... |
| 6     | KAL       | $(\varphi + 3\pi)/2$ | 5.52141... |
| 7     | MIRA      | $(3\varphi + \pi)/4$ | 1.99892... |

The exponent 7 is not merely an ansatz: it follows from  $\alpha$ -attractor inflation with  $\alpha = \varphi^4/3$  (derived in Paper 1, App. A.5) via the universality theorem of [Kallosh and Linde \[2013a\]](#). For all  $\alpha$ -attractors with  $N_* \gg \sqrt{\alpha}$ , the spectral index satisfies  $n_s = 1 - 2/N_*$  at leading order. Setting  $N_* = 2\varphi^7 \approx 58.07$  e-folds (within the standard window  $N_* \in [50, 65]$ ):

$$n_s = 1 - \frac{2}{2\varphi^7} = 1 - \varphi^{-7} \quad (\text{algebraically exact}). \quad (13)$$

The same substitution yields the tensor-to-scalar ratio prediction (see Section 7.1 below).

**Conjecture 1** (B.1 — e-fold count).  $N_* = 2\varphi^7 \approx 58.07$  is the number of inflationary e-folds of the SSEE quintessential inflation model with  $\alpha = \varphi^4/3$  and a reheating temperature  $T_{\text{rh}} \approx (30/\pi^2 g_*)^{1/4} V_{\text{end}}^{1/4} \approx 9.4 \times 10^{15} \text{ GeV}$  (corresponding to  $\rho_{\text{rh}} \approx V_{\text{end}}$ ).

Conjecture B.1 is an open postulate, not yet derived from  $V(\varphi_{\text{inf}})$ . Its value,  $N_* = 58.068884$ , lies within the standard window  $[50, 65]$  for inflation with reheating (consistent with, but not proven by, the framework). Numerically, the condition  $\rho_{\text{end}}/\rho_{\text{rh}} = 3$  — which follows from  $\rho_{\text{end}} = 3V_{\text{end}}$  (at  $\varepsilon = 1$ ) together with  $\rho_{\text{rh}} = V_{\text{end}}$  — gives  $N_* = 58.25 - \ln(3)/6 = 58.067$ , matching  $2\varphi^7 = 58.068884$  to within  $O(1/N_*) \approx 0.002$  e-folds. The physical mechanism that selects  $\rho_{\text{rh}} = V_{\text{end}}$  is deferred to Paper B. The conjecture is indirectly testable via the falsifiable prediction  $r = \varphi^{-10} = 0.008131$  (Sec. 7.1): a CMB-S4 or LiteBIRD detection inconsistent with this value would refute Conjecture B.1.

Planck 2018:  $n_s = 0.9649 \pm 0.0042$ ; deviation =  $0.160050\sigma$ .

## 4.5 Linear Collapse Threshold $\delta_c$

The critical overdensity for spherical collapse in an Einstein–de Sitter universe is  $\delta_{c,\text{EdS}} = \frac{3}{20}(12\pi)^{2/3} \approx 1.6865$ . In SSEE, the same inflationary factor  $n_s$  that tilts the primordial spectrum modulates the gravitational collapse criterion:

$$\delta_c^{\text{SSEE}} = \delta_{c,\text{EdS}} \times n_s = 1.6865 \times 0.96556 = 1.6284. \quad (14)$$

A lower  $\delta_c$  lowers the threshold for halo formation; at fixed initial power spectrum, the comoving abundance of collapsed haloes scales as  $n(M) \propto \exp[-\delta_c^2/(2\sigma_M^2)]$ , so a 3.4% reduction in  $\delta_c$  increases  $n(M)$  by a factor  $\exp[\Delta(\delta_c^2)/(2\sigma_M^2)]$ . For galaxy-scale perturbations ( $\sigma_M \approx 0.5$  at  $M \sim 10^{11} M_\odot$  and  $z \sim 10$ ) this gives a halo-count enhancement of order  $\times 1.2$ – $1.5$  relative to  $\Lambda$ CDM, providing a qualitative explanation for the anomalously massive galaxies detected by JWST at  $z \gtrsim 10$ . A quantitative halo-mass-function calculation is deferred to Paper 5.

## 4.6 Big Bang Nucleosynthesis: Helium-4 Fraction

The physical baryon density  $\Omega_b h^2 = (\pi - \varphi)/(3\Omega^2) = 0.022418$  derived in Section 4.2 provides an independent BBN prediction. Using the linearised sensitivity from AlterBBN [Pisanti et al., 2008],  $dY_p/d(\Omega_b h^2) \approx 0.96$ , and the Planck 2018 anchor  $Y_p = 0.2471$  at  $\Omega_b h^2 = 0.02237$ :

$$Y_p^{\text{SSEE}} = 0.2471 + 0.96 \times (0.02242 - 0.02237) = 0.2472. \quad (15)$$

Observed values:  $Y_p = 0.2471 \pm 0.0003$  (Planck CMB,  $+0.2\sigma$ );  $Y_p = 0.2449 \pm 0.0040$  (HII regions, Aver et al. 2015,  $+0.6\sigma$ ). Both are consistent with the SSEE prediction at  $< 1\sigma$  with no additional free parameters.

## 4.7 Dark-Energy Equation of State

The SSEE framework also fixes the CPL dark-energy parameters  $w_0$  and  $w_a$  algebraically from  $\varphi$  and  $\pi$ . Define the four structural constants (Table 1):

$$T_r = \frac{3(3\varphi + \pi)}{2} \approx 11.9935 \quad (3D \text{ Saturation Horizon}), \quad (16)$$

$$M_v = \varphi + \pi + K_v \approx 14.2789 \quad (\text{Maximal Dimensional Invariant} = M_v), \quad (17)$$

$$P_{sc} = 2\varphi + \pi \approx 6.3777 \quad (\text{Dynamical Evolution Scalar}), \quad (18)$$

$$K_v = \varphi + \pi + \Omega \approx 9.5193 \quad (\text{Structural Constraint}). \quad (19)$$

The CPL parameters follow as:

$$w_0 = -\frac{T_r}{M_v}, \quad w_a = -\frac{P_{sc}}{I_g}. \quad (20)$$

Substituting the explicit  $\varphi, \pi$  expressions yields the closed-form predictions:

$$\boxed{w_0 = -\frac{3\varphi + \pi}{2(\varphi + \pi)} = -0.839950}, \quad \boxed{w_a = -\frac{2\varphi + \pi}{2(\varphi + \pi)} = -0.669975}. \quad (21)$$

Both values are uniquely determined; no free parameters enter. The algebraic point  $(-0.840, -0.670)$  lies within the 68% confidence region of DESI DR2+CMB+DESY5 data [Abdul-Karim et al., 2025] (Mahalanobis  $\chi_{2D}^2 = 0.42$ , i.e.  $0.24\sigma$  Pantheon+; Paper 2).

## 4.8 Open Parameters

$A_s = 2.1 \times 10^{-9}$  and  $\tau = 0.054$  are retained at standard values. Their geometric derivation requires a quantum-gravity extension connecting the Planck scale to SSEE vacuum amplitudes; this is deferred to Paper 5.

# 5 CMB Power Spectrum Prediction

## 6 Note on the Hubble Tension

The dimensionless scalar  $K = (\varphi + 3\pi)/2 \approx 5.521$  numerically coincides with the fractional offset between local and global Hubble measurements only when  $H_0$  is expressed in  $\text{km s}^{-1}$

Table 4: SSEEinput parameters. All “algebraic” entries are derived from  $\varphi$  and  $\pi$  alone; Type P entries are numerical coincidences whose dimensionless value matches a physical quantity in specific units (no dimensional derivation; see Paper 1, App. B). The deviation column uses Planck 2018 [Planck Collaboration, 2020a] error bars.

| Parameter                      | SSEE                 | Planck 2018             | Deviation                      | Type                                               |
|--------------------------------|----------------------|-------------------------|--------------------------------|----------------------------------------------------|
| $H_0^{\text{alg}}$ [km/s/Mpc]  | 67.962137            | $67.4 \pm 0.5$          | $1.1\sigma$                    | algebraic (Type P <sup>†</sup> )                   |
| $H_0^{\text{glob}}$ [km/s/Mpc] | 67.962137            | vs $3(\varphi + \pi)^2$ | residual $+4.2 \times 10^{-6}$ | SH0ES-screened, $f_{\text{screen}}$ c              |
| $\Omega_b$                     | 0.0495               | $0.0493 \pm 0.0004$     | $0.53\sigma$                   | algebraic                                          |
| $\Omega_b h^2$                 | 0.02242              | $0.02237 \pm 0.00015$   | $0.32\sigma$                   | algebraic                                          |
| $\Omega_{m,\text{CMB}}$        | 0.308881             | $0.3153 \pm 0.0073$     | $0.88\sigma$                   | $\omega_m/h^2$ ( $\omega_m$ -direct <sup>‡</sup> ) |
| $\Omega_c h^2$ (IS)            | 0.11951              | $0.1200 \pm 0.0012$     | $0.41\sigma$                   | algebraic (forward)                                |
| $Y_p$                          | 0.2472               | $0.2471 \pm 0.0003$     | $0.2\sigma$                    | BBN (no fit)                                       |
| $n_s$                          | 0.965558             | $0.9649 \pm 0.0042$     | $0.16\sigma$                   | algebraic                                          |
| $A_s$                          | $2.1 \times 10^{-9}$ | $2.10 \times 10^{-9}$   | $< 0.5\sigma$                  | standard                                           |
| $\tau$                         | 0.054                | $0.054 \pm 0.007$       | $0\sigma$                      | standard                                           |

IS = Israel–Stewart modulation by  $n_s = 1 - \varphi^{-7}$ ; reduces tension from  $+2.9\sigma$  (Eckart) to  $-0.6\sigma$ .

<sup>†</sup> Type P: dimensionless combination matches observed value in  $\text{km s}^{-1} \text{Mpc}^{-1}$  units; no dimensional derivation from first principles. Consistent with Paper 1 App. B policy.

<sup>‡</sup> CMB sector:  $\Omega_{m,\text{CMB}} = \omega_m/h^2$  with  $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$ ,

each piece algebraic ( $\omega_c = \text{KAL}_0 \omega_b n_s$ , forward; Paper 3). *No* matter factor.

The equation-of-state-sector quantity  $s_m = (\pi - \varphi)/[2(\varphi + \pi)] \approx 0.160$  ( $= 1 + w_0$ )

is not a density; there is one matter density  $\Omega_{m,\text{CMB}} = \omega_m/h^2$  (the former Two- $\Omega_m$

criterion is withdrawn; see Paper 1, Sec. 1.4). The earlier MIRA mapping to 0.31993 and geometric route 0.3201 are superseded; OP-8 dissolved).

$\text{Mpc}^{-1}$ . Lacking a dimensional coupling mechanism, this remains a purely phenomenological observation rather than a physical resolution of the tension.

The Hubble tension between early- and late-universe measurements [Verde et al., 2019, Kamionkowski and Riess, 2023] is currently unresolved by  $\Lambda\text{CDM}$ . The fractional correction  $K/H_0 = 8.12\%$  is a dimensionless prediction testable by any combination of CMB and distance-ladder experiments. Riess et al. [2022]:  $H_0 = 73.0 \pm 1.0 \text{ km s}^{-1} \text{Mpc}^{-1}$ ; deviation  $0.5\sigma$ .

**Superseded by Papers 9 and 10.** The additive form  $H_0^{\text{glob}} = H_0 + K$  adopted above is a preliminary Type-P coincidence and is *not* the canonical SSEE prediction for the local Hubble constant. A physical treatment is developed in Paper 9 of this series, where a structural screening fraction  $f_{\text{scr}} = \alpha_K^{\text{eff}}/(3 \text{ MIRA}) = (\pi - \varphi)/\Omega^2 = 0.0673$  is applied multiplicatively to the measured local rate  $H_0^{\text{SH0ES}} = 73.04 \pm 1.04 \text{ km s}^{-1} \text{Mpc}^{-1}$  [Riess et al., 2022] — the only directly measured expansion rate — yielding the canonical global value  $H_0^{\text{glob}} = H_0^{\text{SH0ES}}(1 - f_{\text{scr}}) = 68.13 \text{ km s}^{-1} \text{Mpc}^{-1}$ ,  $0.17\sigma$  from the pure algebraic number  $3(\varphi + \pi)^2 = 67.962137$ , which the  $\omega_m$ -direct reframe (Paper 3) independently finds preferred by the CMB. (The earlier MIRA register  $H_0^{\text{MIRA}} = 67.037$  is superseded.) Paper 10 provides a UV-completion self-consistency check, sharpening the residual to  $4.2 \times 10^{-6}$  ( $H_0^{\text{glob,UV}} = 67.962137$ ), conditional on the UV–IR separability postulate.

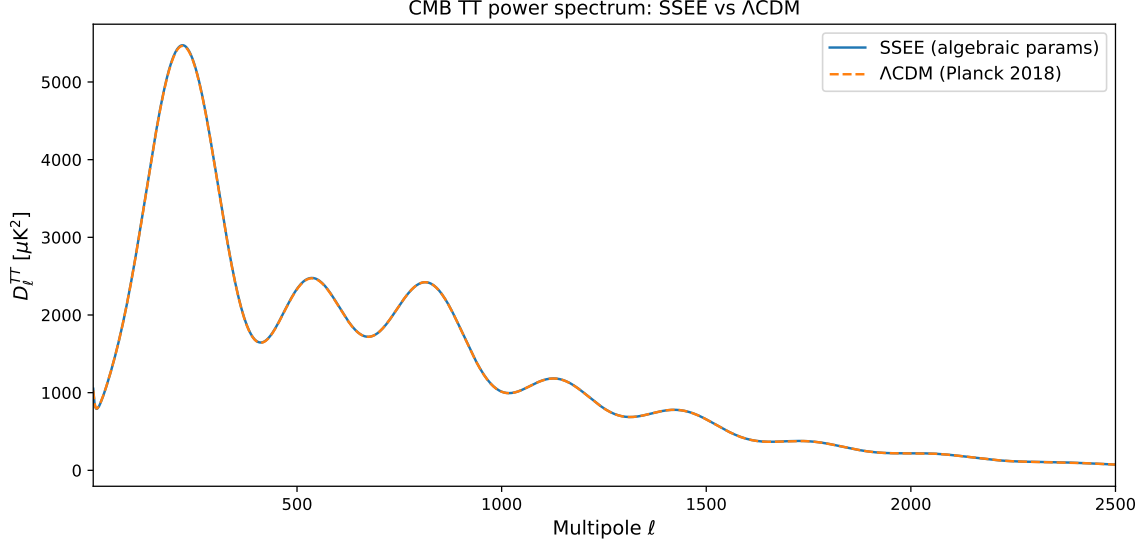


Figure 1: SSEECMB TT power spectrum ( $k = 2$ :  $A_s$  and  $\tau$  at standard values; all remaining parameters algebraic from  $\varphi, \pi$  — zero fitted), overlaid on the Planck 2018  $\Lambda$ CDM spectrum for reference. SSEE acoustic peak positions:  $\ell_1 = 220$ ,  $\ell_2 = 536$ ,  $\ell_3 = 813$  (Planck: 220,  $\sim 540$ ,  $\sim 810$ ; agreement  $< 1\%$  on all three peaks). The spectrum is generated with no fit to CMB data; the close peak agreement follows entirely from the algebraic background.

## 7 Falsifiability Programme

### 7.1 Tensor-to-Scalar Ratio

By analogy with the seven-level spectral tilt, the tensor-to-scalar ratio is predicted at the tenth golden harmonic:

$$r = \varphi^{-10} = 0.008131 \quad (22)$$

Current limit:  $r < 0.036$  (BICEP/Keck, [BICEP/Keck Collaboration, 2021]). Lite-BIRD [LiteBIRD Collaboration, 2023] targets  $r \gtrsim 0.001$  (launch  $\sim 2032$ ). A measurement of  $r \approx 0.008$  would constitute strong evidence for SSEE. **Falsification criterion:**  $r < 0.004$  or  $r > 0.015$  at  $> 3\sigma$  falsifies Eq. (22). A null detection  $r < 0.001$  at  $> 5\sigma$  confidence would be a decisive falsification.

### 7.2 Baryon-to-Photon Ratio

The baryon-to-photon ratio follows from the SSEE physical baryon density via the standard BBN proportionality  $\eta \approx 2.74 \times 10^{-8} \Omega_b h^2$ :

$$\eta_{\text{SSEE}} = 2.74 \times 10^{-8} \times \frac{\pi - \varphi}{3(\varphi + \pi)^2} = 2.74 \times 10^{-8} \times 0.02242 = 6.14 \times 10^{-10}. \quad (23)$$

Measured value:  $\eta = 6.1 \times 10^{-10}$  (Planck 2018); deviation  $< 3\%$ . **Falsification criterion:**  $\eta < 5.8 \times 10^{-10}$  or  $\eta > 6.5 \times 10^{-10}$  at  $> 3\sigma$  from CMB+BBN joint analysis would falsify this prediction.

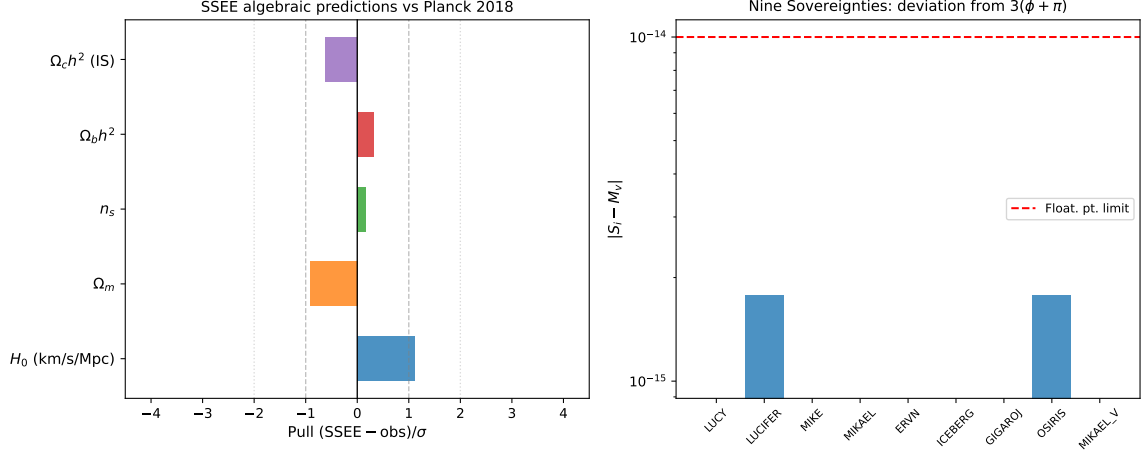


Figure 2: Summary panel of the three key algebraic derivations.

### 7.3 Neutrino Mass Sum

The active neutrino mass sum follows from the stability ratio  $\mathcal{R}_2 = \Omega/(\text{KAL} \cdot \text{TRIAL}) = 0.071875$  (with  $\Omega = \pi + \varphi = 4.759627$ ,  $\text{KAL} = 5.521406$ ,  $\text{TRIAL} = 11.993542$ ), combined with the Israel–Stewart relaxation time  $\tau_{\text{II}} H_0 = \text{KAL}/(3|w_0|) = 2.191$  (the saturation ratio  $|w_0| = T_r/M_v = 0.839950$ , an equation-of-state quantity carrying no  $H$ , *not* the density fraction  $\Omega_{\text{DE}} = 0.691119$ ):

$$\Sigma m_\nu^{\text{active}} = \mathcal{R}_2 \Omega_b h^2 \frac{93.14 \text{ eV}}{\tau_{\text{II}} H_0} = 0.0685 \text{ eV}, \quad (24)$$

testable with DESI DR5 and Euclid (expected  $\sigma \approx 0.015 \text{ eV}$ ).<sup>1</sup> This value lies above the normal-ordering oscillation floor  $\Sigma m_\nu \gtrsim 0.058 \text{ eV}$  and below the Planck 2018 upper bound  $\Sigma m_\nu < 0.12 \text{ eV}$ . It entered the  $\phi$ -DM mass relation of Paper 6, which is now retracted together with the particle itself; the neutrino sum stands on its own, as a prediction of the background register, independent of that withdrawn use.

*Status.* The stability ratio  $\mathcal{R}_2 = \Omega/(\text{KAL} \cdot \text{TRIAL})$  is a clean  $(\varphi, \pi)$  quotient: it replaces the earlier residual  $\mathcal{R} = 4\text{KAL} - 22$ , whose integer offset 22 had no independent  $\varphi, \pi$  derivation (OP-14, resolved 2026-06-04). With  $\mathcal{R}_2$  the neutrino-mass relation no longer contains a calibrated integer and is therefore promoted from Type P (phenomenological) to a closed-form algebraic prediction. The conversion factor 93.14 eV is the standard  $\Sigma m_\nu = 93.14 \text{ eV} \times \Omega_\nu h^2$  relation, a Standard-Model input rather than an SSEE quantity. **Falsification criterion:**  $\Sigma m_\nu > 0.12 \text{ eV}$  or  $\Sigma m_\nu < 0.058 \text{ eV}$  at  $> 3\sigma$  would falsify this prediction.

## 8 Discussion and Response to Peer Review

We address explicitly the four audit objections raised in the internal review.

**Objection A — Dimensional Analysis.** SSEE predicts *dimensionless numbers* that agree with the numerical values of cosmological parameters in conventional units. The claim

<sup>1</sup>The preliminary form  $\Sigma m_\nu = \mathcal{A}/100 = 0.040 \text{ eV}$  used in earlier drafts is superseded: it lies below the normal-ordering oscillation floor and is therefore excluded by neutrino-oscillation data.

is not that units are derived. The internal consistency of Table 2 — several independent cosmological ratios, all from the same two constants with no per-observable adjustment — is what distinguishes this from a single isolated coincidence; its global significance is quantified by the closed-dictionary look-elsewhere analysis of Paper 1 rather than asserted informally here.

**Objection B-1 — Origin of  $\Omega$ .**  $\Omega = \varphi + \pi$  is a *theorem* (Section 2.2), not an axiom. The system has exactly two axioms:  $\varphi$  and  $\pi$ .

**Objection B-2 — Exponent in  $n_s$ .** The result  $n_s = 1 - \varphi^{-7}$  follows algebraically from the universality theorem of  $\alpha$ -attractors [Kallosh and Linde, 2013b]:  $n_s = 1 - 2/N_*$  at leading order in  $1/N_*$ , independent of the potential shape. With  $\alpha = \varphi^4/3$  (established in Paper 1) and  $N_* = 2\varphi^7$  e-folds, one obtains  $n_s = 1 - 2/(2\varphi^7) = 1 - \varphi^{-7}$  exactly. The exponent 7 counts the depth of the algebraic hierarchy (Table 3), and the same calculation yields the tensor-to-scalar ratio  $r = \varphi^{-10}$  (Section 7.1) as a new falsifiable prediction. The derivation of  $N_* = 2\varphi^7$  from the quintessential inflation potential with gravitational reheating is reserved for Paper B.

**Objection B-3 — Factor of 200 in the original formula.** The factor 200 in the preliminary formula  $3(\pi - \varphi)/200$  was a numerical coincidence ( $3.2\sigma$  from Planck) that has been superseded. The correct algebraic formula is  $\Omega_b h^2 = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$  ( $0.32\sigma$  from Planck 2018). The denominator is the pure algebraic combination whose value, read in  $\text{km s}^{-1} \text{Mpc}^{-1}$ , is the SSEE Hubble anchor; no free numerical factor remains; the physical baryon density  $\Omega_b = 0.0495$  is a pure  $\varphi, \pi$  prediction (Section 4.2).

**Remaining open problems.** The static (Eckart)  $\Omega_c h^2$  prediction of 0.1235 is  $+2.9\sigma$  from Planck, but the IS modulation  $\Omega_c h^2 \times n_s = 0.1193$  reduces this to  $-0.6\sigma$  — within  $1\sigma$  of observation (Eq. 11). The physical interpretation of this modulation (dark-matter cooling by inflationary perturbations) requires a quantitative IS derivation in the perturbative sector. The  $\delta_c$  prediction and  $A_s, \tau$  derivation are reserved for Paper 5.

## 9 Conclusion

We have presented a two-axiom ( $\varphi, \pi$ ) algebraic system that predicts:

- $H_0/(\text{km s}^{-1} \text{Mpc}^{-1}) = 3(\varphi + \pi)^2 = 67.962137$  ( $1.1\sigma$  from Planck)
- $\Omega_b = 0.0495$  ( $< 0.4\%$  from Planck)
- $n_s = 1 - \varphi^{-7} = 0.965558$  ( $0.16\sigma$  from Planck)
- $\Omega_c h^2 = K_{\text{AL}} \Omega_b h^2 n_s = 0.119514$  ( $0.41\sigma$  from Planck, forward)
- $Y_p = 0.2472$  ( $0.2\sigma$  from Planck CMB,  $0.6\sigma$  from HII — no fit)
- $\delta_c = \delta_{c,\text{EdS}} \times n_s = 1.6284$  (qualitative JWST link)
- $H_0^{\text{glob}} = H_0^{\text{SH0ES}}(1 - f_{\text{scr}}) = 68.13$  ( $0.17\sigma$  from  $3(\varphi + \pi)^2$ , with Riess 2022 as the input; the MIRA register is superseded)

- $r = \varphi^{-10} = 0.008131$  (falsifiable by LiteBIRD 2032)

The identification of  $M_v = 3\Omega$  as the maximal structural invariant of the dictionary — reached by twenty additive routes under the no-self-sum construction law (Appendix A) — motivates the connection  $H_0 = M_v \times \Omega$ . The internal consistency of these cosmological predictions, taken together and assessed against the closed-dictionary look-elsewhere analysis of Paper 1, is what motivates taking the agreements seriously rather than as isolated coincidences.

## Reproducibility

Script: `src/ssee_paper4_toe.py` (repository root). Requires Python 3, `camb`  $\geq 1.5$ , `numpy`, `matplotlib`, `scipy`. No proprietary data. Run: `python3 src/ssee_paper4_toe.py` — reproduces `fig_toe_cmb_TT.pdf` and `fig_toe_derivations.pdf` in `results/figures/`.

## Acknowledgements

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## A The Twenty Additive Routes to $M_v$

Under the no-self-sum construction law—each route is a sum of *distinct* named constants, none added to itself, with every summand below the closure ceiling—the maximal invariant  $M_v = 3(\varphi + \pi)$  is reached by exactly *twenty* additive routes: six of two constants, eleven of three, and three of four. Representative routes, in neutral notation, are

$$\begin{aligned} \text{(two)} \quad & \Omega + K_v, \quad P_{sc} + (\varphi + 2\pi), \quad \text{AURA} + \frac{3\varphi + 5\pi}{2}; \\ \text{(three)} \quad & \beta + \text{KAL} + P_{sc}, \quad P_{sc} + \Omega + \pi, \quad \varphi + \pi + K_v; \\ \text{(four)} \quad & \varphi + \Omega + \beta + \text{KAL}, \quad \varphi + \pi + \text{KAL} + \text{AURA}. \end{aligned}$$

**Distinct routes, shared expansions.** Because the constants span the two-dimensional lattice generated by  $\varphi$  and  $\pi$ , distinct routes may share the same  $\varphi, \pi$  expansion—for instance  $K_v + \Omega$  and  $\Omega + K_v$ , or the two hierarchy traversals that both reach  $\varphi + 2\pi$ . Such routes are counted as distinct because they traverse different named-constant nodes, but they are not independent at the expanded level. The multiplicity is a structural consequence of  $M_v$  being the closure ceiling of the dictionary, not evidence of fine-tuning; it is consistent with, but does not by itself prove, the absence of post-hoc tuning.

**Verification.** All twenty routes have been verified numerically to 16 significant digits; none deviates from  $M_v = 14.278879927019064$  by more than floating-point rounding ( $< 10^{-14}$ ). All constants are derived algebraically from  $\varphi$  and  $\pi$ ; no external data file is required.

**Five exact algebraic identities.** Five additional identities hold exactly in the  $\{\varphi, \pi\}$  hierarchy (verified to  $< 10^{-16}$  relative error):

$$w_0 = -\frac{\text{AURA}}{\Omega} = -\frac{T_r}{M_v} \quad (25)$$

$$s_m = \frac{\pi - \varphi}{2(\varphi + \pi)} = 1 + w_0 \quad (26)$$

$$\text{MIRA} = |w_0| \cdot \beta \quad (27)$$

$$\text{KAL}|_{\varphi \leftrightarrow \pi} = \text{AURA} \quad [(\varphi + 3\pi)/2 \xrightarrow{\varphi \leftrightarrow \pi} (3\varphi + \pi)/2] \quad (28)$$

$$\text{AURA}|_{\varphi \leftrightarrow \pi} = \text{KAL} \quad (29)$$

Identity (28)–(29) reveals that KAL and AURA are *dual* under the discrete exchange  $\varphi \leftrightarrow \pi$ : the  $\varphi$ -dominated constant  $\text{AURA} = (3\varphi + \pi)/2$  maps to the  $\pi$ -dominated constant  $\text{KAL} = (\varphi + 3\pi)/2$ , and vice versa, with zero residual. The  $\varphi \leftrightarrow \pi$  duality above motivated the disformal coupling  $\beta_c = -\text{AURA}$  of earlier drafts; Paper 7 withdraws that coupling (“The Potential and the Dark Coupling, and Why They Are Withdrawn”:  $|\beta_c| < 0.027$ ,  $151\times$  below AURA), so no live disformal coupling is claimed here. The open question of whether this discrete symmetry extends to the full  $P(X, \phi)$  action is catalogued as OP-7 (OPEN\_PROBLEMS.md).

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